

Radix Sort

input array: 170, 45, 75, 90, 802, 24, 2, 66, 473, 1001

step 1: sort according to 1st digit.

170, 90, 1001, 802, 02, 473, 24, 45, 75, 66

step 2: sort according to 2nd digit.

1001, 802, 002, 024, 045, 066, 170, 473, 075, 090

step 3: sort according to 3rd digit.

1001, 002, 024, 045, 066, ~~090, 170~~, 075, 090, 170, 473, 802

step 4: sort according to 4th digit.

2, 24, 45, 66, 75, 90, 170, 473, 802, 1001

Properties:

i) Not Subtract & Conquer Neither Divide & Conquer.

(Not Subtraction)

ii) External or outplace.

iv) Data dependant.

v) Best & worst case: $O(nk)$

$\hookrightarrow$ no. of digits.

Stable.

6zTerndU